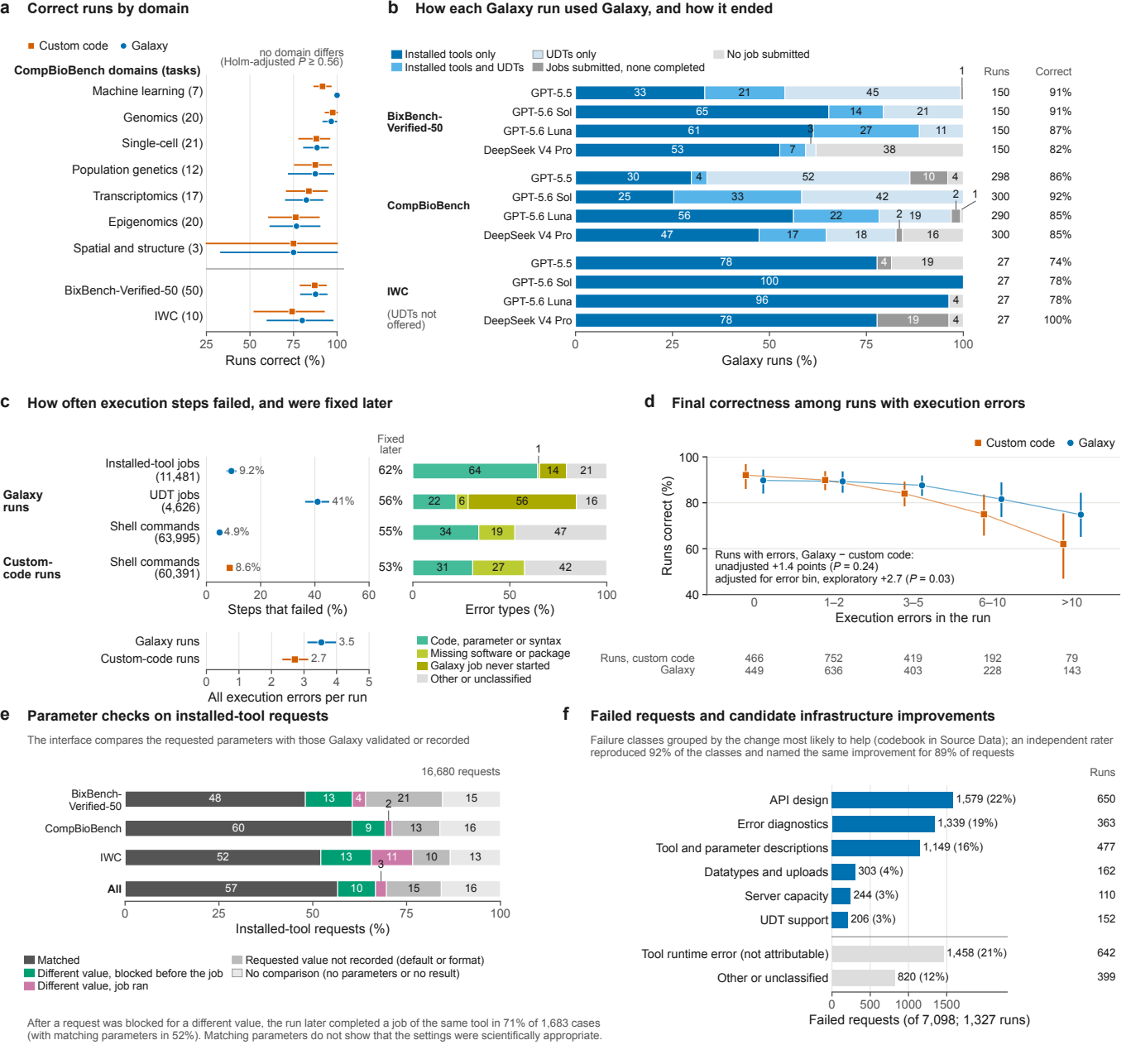

a
Correct runs by domain
Custom code
Galaxy
no domain differs
(Holm-adjusted P ≥ 0.56)
CompBioBench domains (tasks)
Machine learning (7)
Genomics (20)
Single-cell (21)
Population genetics (12)
Transcriptomics (17)
Epigenomics (20)
Spatial and structure (3)
BixBench-Verified-50 (50)
IWC (10)
25
50
75
100
Runs correct (%)
b
How each Galaxy run used Galaxy, and how it ended
Installed tools only
Installed tools and UDTs
UDTs only
Jobs submitted, none completed
No job submitted
Runs
Correct
BixBench-Verified-50
GPT-5.5 33 21 45 150 91%
GPT-5.6 Sol 65 14 21 150 91%
GPT-5.6 Luna 61 3 27 11 150 87%
DeepSeek V4 Pro 53 7 38 150 82%
CompBioBench
GPT-5.5 30 4 52 10 4 298 86%
GPT-5.6 Sol 25 33 42 2 1 300 92%
GPT-5.6 Luna 56 22 2 19 290 85%
DeepSeek V4 Pro 47 17 18 16 300 85%
IWC
(UDTs not offered)
GPT-5.5 78 4 19 27 74%
GPT-5.6 Sol 100 27 78%
GPT-5.6 Luna 96 4 27 78%
DeepSeek V4 Pro 78 19 4 27 100%
0
25
50
75
100
Galaxy runs (%)
c
How often execution steps failed, and were fixed later
Fixed later
Galaxy runs
Custom-code runs
Installed-tool jobs (11,481) 9.2% 62% 64 14 21
UDT jobs (4,626) 41% 56% 22 6 56 16
Shell commands (63,995) 4.9% 55% 34 19 47
Shell commands (60,391) 8.6% 53% 31 27 42
0
20
40
60
Steps that failed (%)
0
50
100
Error types (%)
Galaxy runs 3.5
Custom-code runs 2.7
0
1
2
3
4
5
All execution errors per run
Code, parameter or syntax
Missing software or package
Galaxy job never started
Other or unclassified
d
Final correctness among runs with execution errors
Custom code
Galaxy
100
80
60
40
Runs correct (%)
Runs with errors, Galaxy − custom code:
unadjusted +1.4 points (P = 0.24)
adjusted for error bin, exploratory +2.7 (P = 0.03)
0
1–2
3–5
6–10
>10
Execution errors in the run
Runs, custom code 466 752 419 192 79
Galaxy 449 636 403 228 143
e
Parameter checks on installed-tool requests
The interface compares the requested parameters with those Galaxy validated or recorded
16,680 requests
BixBench-Verified-50 48 13 4 2 21 15
CompBioBench 60 9 13 16
IWC 52 13 11 3 10 13
All 57 10 15 16
0
25
50
75
100
Installed-tool requests (%)
Matched
Different value, blocked before the job
Different value, job ran
Requested value not recorded (default or format)
No comparison (no parameters or no result)
After a request was blocked for a different value, the run later completed a job of the same tool in 71% of 1,683 cases (with matching parameters in 52%). Matching parameters do not show that the settings were scientifically appropriate.
f
Failed requests and candidate infrastructure improvements
Failure classes grouped by the change most likely to help (codebook in Source Data); an independent rater reproduced 92% of the classes and named the same improvement for 89% of requests
Runs
API design 1,579 (22%) 650
Error diagnostics 1,339 (19%) 363
Tool and parameter descriptions 1,149 (16%) 477
Datatypes and uploads 303 (4%) 162
Server capacity 244 (3%) 110
UDT support 206 (3%) 152
Tool runtime error (not attributable) 1,458 (21%) 642
Other or unclassified 820 (12%) 399
0
500
1000
1500
Failed requests (of 7,098; 1,327 runs)